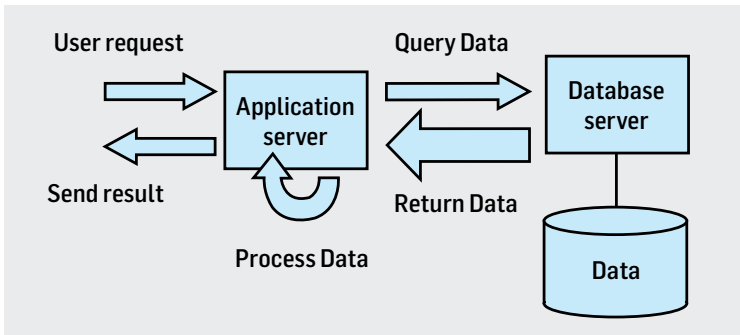


Traditional Approach of Data Storage and Usage

Traditional data systems, such as data warehouses and relational databases, have been around since more than four decades. They have been the primary method for enterprises to store and analyze their data. Regardless of the existence of other data storage technologies, the major chunk of enterprise data can be found in these traditional systems. However, traditional systems were always designed from the ground up to work with only structured data. With the passage of time, the enterprises needed to store more and more detailed information for longer periods of time, mostly in areas such as Health and Finance. Managing the volume and cost of this data growth within these traditional systems is usually a stress point for IT organizations.



SOURCE: IBM CORPORATION

Traditional approach: Move data to program

ACID (Atomicity, Consistency, Isolation, Durability) based systems and the ideology around them are still undeniable for running the business. These systems took a long time to build and support business decisions that run some of the enterprises today. Traditional systems can store around petabytes (PB) of data. However, these systems were not designed to solve a number of today's data challenges. The speed, complexity, and cost of using these traditional systems to address these new data challenges would be extremely high. Unstructured data isn't organized but contains tags, markers, or some method for organizing the data. Examples of unstructured data include social media data structures (Twitter, Facebook), RFID, GPS coordinates, machine sensors, and so on. The volume of this unstructured data, generated on a daily basis, is enormous compared to the structured data. Moreover, this unstructured data is just as critical as the structured internal data being stored in relational databases.